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SOIT
RESEARCH + INNOVATION
NEWSLETTER

Research. Innovation. Impact.

School of Information Technology | STADIO Higher Education

Inside this issue

RESEARCH OUTPUT SNAPSHOT

AS AT 6 SEPTEMBER 2026 | TRACKED SOIT SUBMISSIONS AND OUTPUTS

21
Tracked records

15
Conference records

6
Journal records

12
Accepted outputs

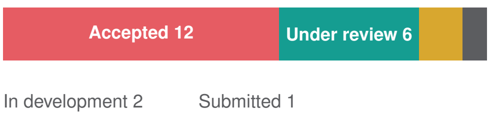
From activity to impact

SOIT guidance favours a quality-first trajectory: credible peer-reviewed outputs, evidence-ready publication records and a sustained 2-3 year pathway toward stronger journals and collaboration.

PIPELINE STATUS

57% accepted. Another 29% is under review.

The active portfolio spans conference and journal routes, with visible clusters in student success, AI, cybersecurity, gamification, AutoML and applied analytics.



What the numbers say

- **7 staff contributors** are represented in the active reporting dataset.
- **10 conference outputs + 2 journal outputs** are recorded as accepted.
- **6 additional outputs** are under review, strengthening the near-term pipeline.
- Recurring themes include **student retention**, **AI in higher education**, and **AI/cybersecurity**.

Operational priority

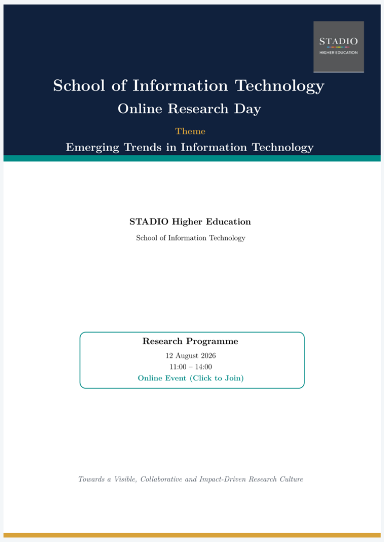
Turn every accepted output into a complete evidence pack: acceptance, peer-review evidence, proceedings or journal details, invoices/fees where applicable, and ethics/gatekeeper documentation.

Quality signal

The objective is not simply more outputs; it is a reliable research pipeline in which each output is visible, defensible and claim-ready.

RESEARCH DAY WEBINAR RECAP

12 AUGUST 2026 | EMERGING TRENDS IN INFORMATION TECHNOLOGY



A VISIBLE, COLLABORATIVE AND IMPACT-DRIVEN RESEARCH CULTURE

Three hours. One research community. Seven research-facing sessions.

The SOIT Online Research Day created a focused platform for staff, postgraduate researchers and invited scholars to share emerging work, receive scholarly feedback, identify publication opportunities and strengthen collaboration across the School.

PROGRAMME AT A GLANCE

11:00	**Opening + welcome**	Dr Michael Ajayi: Research Day purpose and objectives.
11:05	**Keynote**	Dr Tshepo Kukuni: Emerging Trends in Information Technology Research.
11:30-13:10	**Six research talks**	Scientific computing, cybersecurity, AutoML, recruitment analytics, at-risk analytics and digital health.
13:10	**Research development**	Dr Belinda Ndlovu: practical writing for journals, conferences and proposals.
13:40	**Collaboration discussion**	Publication pathways, mentorship and the emerging SOIT research agenda.
13:55	**Closing**	Dr Jan Mentz, Head of School.

AI + Scientific Computing

AutoML, physics-informed neural networks, algorithm configuration and generation.

Cybersecurity + Digital Trust

Blockchain-enabled DDoS mitigation, SDN, resilient systems and network security.

Education + Analytics

Recruitment analytics, at-risk identification, learning analytics and research skills development.

Next move from the webinar

Convert conversations into concrete writing groups, co-authorship, supervision links and publication-ready manuscripts. The event's value is realised when visibility becomes output.

INNOVATION

ASSESSMENT DESIGN IN THE AI ERA

The design question is no longer “How do we stop AI?” It is “How do we require credible evidence of learning when AI is available?”

A FOUR-STAGE ASSESSMENT ARCHITECTURE

1. FOUNDATIONS

Individual, time-bounded checkpoints on concepts and core skills. Limit or exclude AI where foundational competence must be independently demonstrated.

→

2. BUILD

Authentic project work with explicit AI rules. Students design, integrate, test, iterate and document decisions.

→

3. TRACE

Require evidence trails: commits, test cases, version history, prompt/decision logs, citations and debugging records.

→

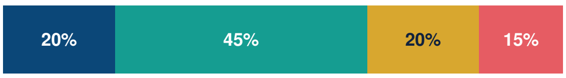
4. DEFEND

Verify ownership through live demonstration, oral defence, code walkthrough, debugging or short transfer tasks.

Five design rules

- Assess **process + product**, not only the final artefact.
- Make AI use **explicit, bounded and disclosable**.
- Protect fundamentals with **individual checkpoints**.
- Require **testing, reproducibility and evidence trails**.
- Make students **explain and transfer** what they built.

ILLUSTRATIVE WEIGHTING - NOT POLICY



Foundation check	Individual concepts / fundamentals
Guided project	Authentic build with bounded AI use
Controlled validation	Unseen task, test or debugging
Viva + reflection	Explain, justify and defend decisions

Why this belongs in SOIT

The School’s 2026-2028 agenda explicitly includes technology-enhanced pedagogy, digital curriculum design, AI-driven assessment and personalised learning. The proposed research lab also identifies AI-assisted programming education and AI dependency as research themes.

What this could look like in a programming module

Concept check → **GitHub-based guided project** → **test cases + commit evidence** → **live code walkthrough / debugging task**. The final mark is supported by multiple forms of evidence rather than a single finished artefact.

SOIT RESEARCH AGENDA

FOCUS AREAS 2026-2028 | APPLIED, SOCIALLY RELEVANT, IMPACT-DRIVEN

The School's research agenda is organised around three complementary areas aligned to STADIO's RRITE values: **Relevant, Responsive, Innovative, Transformative and Effective**.

01

Applied Artificial Intelligence for Industry and Society

Responsible and explainable AI; AI for social good; automation and robotics; AI-enabled productivity and innovation for SMEs.

02

Use of ICT in Education and Digital Innovation

Technology-enhanced pedagogy; digital curriculum; AI-driven assessment; personalised learning; virtual labs; micro-credentials; digital inclusion.

03

Data-Driven Decision Support and Analytics

Decision-support systems; predictive analytics; dashboards; visualisation; policy and risk models; data ethics, privacy and governance.

WHAT IS ALREADY VISIBLE IN THE 2026 PIPELINE

- Student retention + early warning
- AI in higher education
- AI + cybersecurity
- AutoML + PINNs
- Gamification
- UX + applied information systems

From focus area to research output

The strongest opportunity is to connect existing teaching, supervision and applied projects to one of the three focus areas, then move each project through a disciplined research pipeline: clear question, appropriate ethics/permission, credible method, peer-reviewed output and an evidence-ready record.

Research governance reminder

Ethics approval and institutional/gatekeeper permission are distinct. Where STADIO people, data or systems are involved, obtain the appropriate approvals **before** data collection.

Quality over volume

Target credible peer-reviewed outputs, avoid predatory outlets, and keep a clean evidence trail from proposal to publication and claim.

INDUSTRY ECOSYSTEM SPOTLIGHT

SAICTA | CONNECTING EDUCATION, TALENT AND SOUTH AFRICA'S ICT ECOSYSTEM

Potential SOIT engagement

1. Define an institutional engagement pathway.
2. Use events and masterclasses for curriculum and staff-development intelligence.
3. Link students and alumni to Industry Connect opportunities.
4. Position SOIT research, prototypes and thought leadership in relevant channels.
5. Explore speakers, challenge statements, datasets and applied research collaboration.

WHY SAICTA MATTERS TO SOIT

A potential bridge between academic capability and the ICT ecosystem.

SAICTA positions itself as a national ICT association connecting companies and professionals across sectors, with emphasis on professional development, networking, industry visibility and policy influence. For SOIT, the value proposition is not a logo on a page - it is access to conversations, opportunities and industry signals.

Education ↔ Industry

Talent ↔ Opportunity

Innovation ↔ Impact

SAICTA platforms relevant to a School of IT

- **Industry Connect**: a channel linking ICT employers and job seekers.
- **Professional development**: training and CPD-oriented activities.
- **Digital Dawn + Masterclasses**: trend, networking and industry-insight sessions.
- **ICT Excellence Awards**: visibility for innovation and leadership.
- **Thought leadership + policy advocacy**: a broader sector voice.

Strategic fit

SAICTA's emphasis on talent, networking, professional development and ICT-sector visibility aligns naturally with SOIT's goals for employability, partnerships, research visibility and digital innovation.

VISIT SAICTA.ORG

VISION + FUTURE MILESTONES

SOIT APPLIED AI, ANALYTICS AND DIGITAL INNOVATION LAB

LAB VISION

To establish a leading Applied AI, Analytics and Digital Innovation Lab that advances research, innovation, student success and industry collaboration through evidence-based computing solutions.

FOUR RESEARCH PILLARS

P1	Learning Analytics, Student Success, Educational AI
P2	Human-Centred AI, Inclusive Design, Accessibility
P3	Career Intelligence, Employability Analytics, Recommendation Systems
P4	Cybersecurity, Data Science, Digital Innovation

PROPOSED INNOVATION PROJECTS

AI Academic Advisor
Student risk monitoring, intervention recommendations and academic planning.

Document Fluency AI
Policy interpretation, instructional guidance and intelligent decision support.

AI Research Assistant
Literature analysis, proposal generation and research-gap identification.

Assessment Design in the AI Era
AI-resilient assessment architecture combining foundations, authentic build, evidence trails and controlled defence of learning.

2027-2030 OUTPUT AMBITION

Journal papers	**15+**
Conference papers	**20+**
Honours projects	**25+**
Masters studies	**10+**
Industry partnerships	**5+**
Software prototypes	**8+**

PROPOSED MILESTONE PATHWAY

2026 Foundation — Governance + flagship studies → **2027** Launch — Clusters + student projects → **2028** Scale — Cross-campus + showcase → **2029** Translate — Masters + industry pilots → **2030** Impact — Consolidate + next cycle

What success would look like

A visible platform where staff research, postgraduate supervision, student innovation and industry collaboration reinforce one another - producing publications, prototypes, talent and measurable institutional impact.

2027 RESEARCH LAUNCH YEAR

SOIT RESEARCH CALENDAR | PROPOSED MONTHLY TIMELINE

LAUNCH-YEAR INTENT

Build a predictable annual research rhythm that moves ideas into manuscripts, conference submissions, collaborative projects and visible School-level research outputs.

MONTHLY RESEARCH CALENDAR

JAN	Research launch + output mapping
FEB	Research Writing Retreat
MAR	SACLA, SAICSIT 2027 prep: manuscript development
APR	
MAY	SOIT Webinar 1
JUN	Mid-year pipeline review
JUL	STADIO Conference 2027 prep: manuscript development
AUG	SOIT Research Day
SEP	SOIT Webinar 2
OCT	Research Lecture and Project Day
NOV	Publications + claim sprint
DEC	Annual review + 2028 pipeline

ANCHOR ACTIVITIES

Writing Retreat
Concentrated manuscript development and writing-group formation.

Webinar 1 + 2
Research methods, ethics, publication development and invited expertise.

Research Day
Internal showcase, feedback, collaboration and publication conversion.

Conference Prep
SACLA 2027, SAICSIT 2027 and STADIO conference submission readiness.

Research rhythm
Write → Share → Submit → Showcase → Translate

Launch-year success measure

Each major activity should leave a traceable output: a manuscript draft, ethics submission, conference paper, collaboration, prototype, publication record or claim-ready evidence pack.

Planning note: the 2027 calendar is an indicative SOIT research-planning framework for the launch year. Specific webinar, retreat and conference dates should be confirmed once institutional and conference calendars are published.

THE NEXT MOVE

TURN RESEARCH ACTIVITY INTO VISIBLE IMPACT.

The opportunity for SOIT is not only to produce more research. It is to build a repeatable system that connects teaching innovation, credible research design, industry relevance, student development and visible outputs.

Near-term priorities

- Research launch year (2027).
- Move accepted and under-review work through publication and claim pipelines.
- Convert Research Day connections into writing groups, co-authorship and mentorship.
- Pilot AI-resilient assessment designs that generate evidence of genuine learning.
- Translate the Research Lab proposal into a staged implementation roadmap.
- Define high-value industry engagement channels, including potential SAICTA opportunities.

Research. Innovation. Impact.

Credit: Dr Jan Mentz | SOIT - Head of School

Editor: Dr Michael Ajayi | SOIT Research Representative

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Sources used in this issue: SOIT Research Reporting Dashboard - 2026; Research at STADIO: What is Achievable, Fundable & Rewarded (6 Sep 2026); SOIT Online Research Day Programme (12 Aug 2026); SOIT Research Focus Areas 2026-2028; SOIT Applied AI, Analytics and Digital Innovation Lab Proposal (12 Aug 2026); SAICTA official website (About Us, Membership, Events, Industry Connect), accessed 6 Sep 2026. The 2027 Research Launch Year calendar is an editorial planning framework and not an approved institutional calendar.